

A new approach for non-contact, automated measurement of respiration rate using structured light

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Abstract—The respiration rate is one of the essential indicators of human health. Therefore, respiration rates can be used to diagnose several diseases, such as pneumonia. At medical facilities, breaths are measured by a respiration belt or manually counted by medical professionals. However, during the influenza season, hospitals and medical professionals are overloaded and easy, rapid and reliable respiration rate measurement is essential. In this study, we proposed to measure respiration rate utilizing a generated structured light pattern in 10 seconds. We used a simple web camera and mini projector. There are 18 participants (5–45 years old) were enrolled in this study. As a result, the correlation coefficient between the actual values and the experimental values was 0.91. The proposed method appears promising for non-contact, rapid, reliable respiration rate measurement.

Keywords—Respiration rate, structured light, non-contact respiration rate measurement

I. INTRODUCTION

Respiration rate is one of the essential vital signs to monitor human health [1] and is an indicator of serious illness. Such as cardiac arrest, pneumonia, and lung diseases increase the breath count of patients, and those with higher respiratory rates show a higher risk of mortality in patients [2]. At medical facilities, the respiratory rate is measured by a respiratory belt and the contact type, and doctors or nurses manually count patients' chest movements [3].

When breathing in, the diaphragm and external intercostal muscles contract, causing the diaphragm to move downward and the rib cage to lift. During exhalation, the diaphragm moves upward, causing the chest cavity to decrease in size. At the same time, the intercostal muscles, which are situated between the ribs, relax, allowing the rib cage to move downward and inward. This coordinated relaxation of the diaphragm and intercostal muscles results in the expulsion of air from the lungs., leading to a decrease in the size of the rib cage and the expulsion of air in the lungs: the diaphragm moves upward, and the rib cage lowers [4]. During influenza season or winter, hospitals are overloaded, and medical professionals are over-busy. Especially in developing countries, where patients are not equipped with enough medical devices, they wait long hours for medical services.

Contactless radars are used widely to detect respiratory rates. However, body movement affects measurement quality, and researchers are working on body movement cancellation algorithms. Then, especially for children, we need a more stable method to get respiratory rate [5].

Broad methods use contact-type sensors for breath number measurements [6]. Also, some researchers use simple cameras and photoplethysmography to measure respiratory rate. However, lightning and movement affect measurement quality [7–9].

To reduce this disadvantage and help medical professionals, we proposed developing a new approach for non-contact, automated respiration rate measurement utilizing structured light. We used a simple web camera, mini projector and laptop for signal and image processing to calculate respiration rate. A structured light pattern shown in Fig.1 was projected to the participant's chest, and the web camera recorded chest movement for 10 seconds. We used MATLAB software to process the video and calculate the respiration rate.

Section II (A) explains the design concept of the respiration rate measurement approach, and section II (B, C) describes an experimental procedure and software prototype for system evaluation. Section III shows the measurement results and provides a discussion; concluding remarks are given in Section IV.

II. MATERIALS AND METHODS

A. Design concept of the respiration rate measurement approach

The prototype of the proposed method is shown in Fig.1. The mini projector, a Samsung SP-H03 Pico Projector, a web camera(Logitech C270 HD Webcam) and MATLAB software installed on the Lenovo Thinkpad X1 laptop were used to implement the proposed method. The structured light generated by MATLAB software is shown in Fig.2. Specifically, we created 24 non-overlapping dot patterns using combinations of RGB colours.

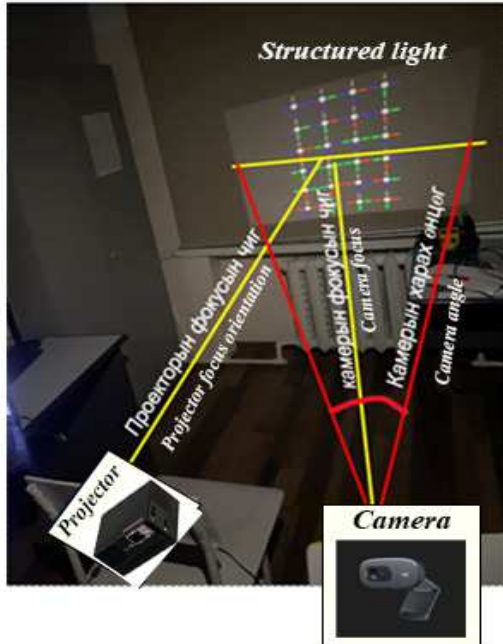


Fig. 1 Measurement prototype

Structured light is a technique used in computer vision and optical metrology to measure the shape of an object or to capture three-dimensional data. The technique is valued for its precision and capability to capture fine details, making it a powerful tool in many fields that require accurate 3D measurements. It involves projecting a known pattern of light (often grids, stripes, or dots) onto a surface and observing how this pattern is deformed by the surface's geometry [10].

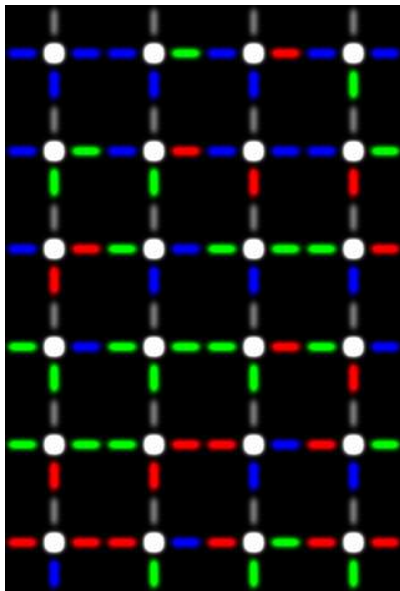


Fig. 2. Generated structured light pattern

The algorithm was implemented according to the block diagram shown below(Fig. 3).

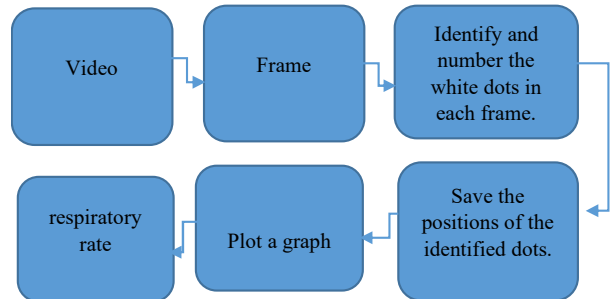


Fig. 3. Algorithm block diagram

i. Create frames from the video.

The structured light pattern projected onto the subject's chest was captured using a video camera for a duration of 10 seconds. A total of 150 frames were extracted from the video using MATLAB. The appearance of one frame is depicted in Fig. 4.

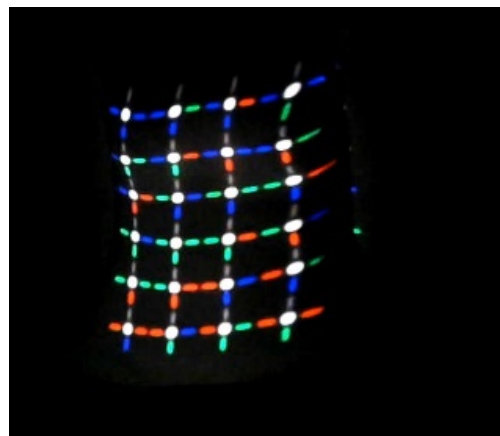


Fig. 4. Extracted single frame from video

ii. White dot detection

Firstly, the position of a white circle on each frame was detected. First, a white circle is identified on the 150 frames generated over 10 seconds. The RGB image is converted into a grayscale image to determine this circle, and a threshold value is set for the converted image. Pixels exceeding this threshold value are identified as part of the white circle. This process uses the MATLAB functions `rgb2gray` (to convert the RGB image to a grayscale image), `regionprops`, and `circles`.

Figure 5 shows detected dots with blue circles on each frame.

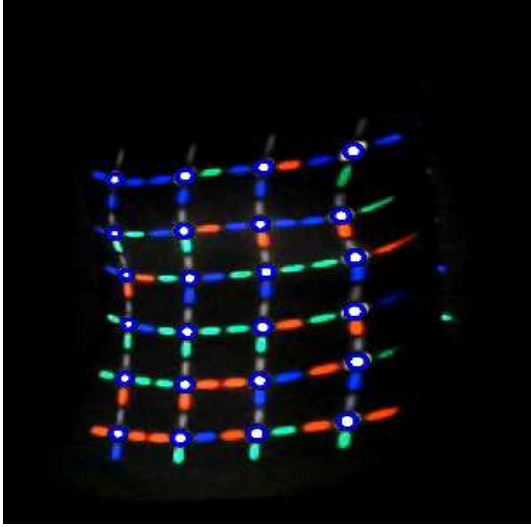


Fig. 5. Detected white dots

iii. White dots indexing

After detecting the white dots, those dots are numbered from 1 to 24 to locate the position of each dot, as shown in Figure 6.

This point is defined using the Centroid function in the MATLAB program and sorted with the “sort” function. The points are in the same order for all frames generated from the video.

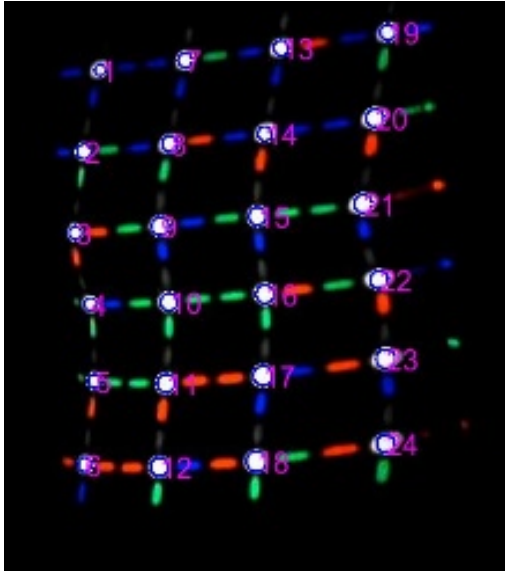


Fig. 6. Indexed dots

B. Experimental procedures

The experimental setup is illustrated in Figure 7. Structured light is projected onto the chest area of a human subject, with the projector positioned perpendicular to the subject. The camera is arranged to capture the entire structured light pattern. Movements of the human chest were recorded for up to 10 seconds.

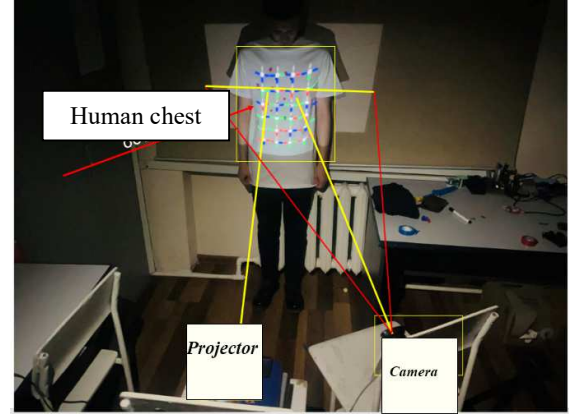
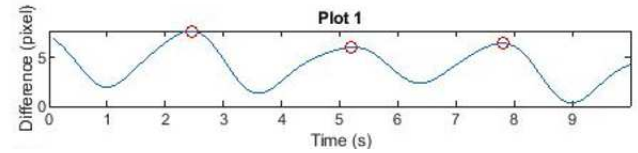


Fig. 7. Experimental setup

C. Respiration rate calculation

We located each dot on frame and tracked the white dot's location in pixels. From pixel location movements, we created



the respiration signals that are shown in Fig.8.

Fig. 8. Respiration signal of single frame

From the above respiration signal we detected number of peaks. From peaks we calculated respiration rate in 60 seconds by Eq.1.

$$\text{Breathes per minute} = \text{number of peaks} \times 6 \quad (1)$$

III. RESULTS AND DISCUSSION

The chest movements of 18 individuals aged 5-45 were recorded, and we manually counted their breaths using an automatic counter to evaluate our new approach. We illuminated structured light on the participants' chests and recorded video for 10 seconds. Analyzing the 24 dots position, we used only four points №10, №11, №16, and №17 to determine the number of breaths. These four points were chosen because the abdominal area generally moves during breathing. The graph generated based on the positions of 24 points during a 10-second breathing of a five-year-old child is shown in Fig.9.

Peak values are detected using a signal processing tool on MATLAB. By selecting points 10, 11, 16, and 17 within the experimental scope, the number of breaths was determined using the average value of these four points.

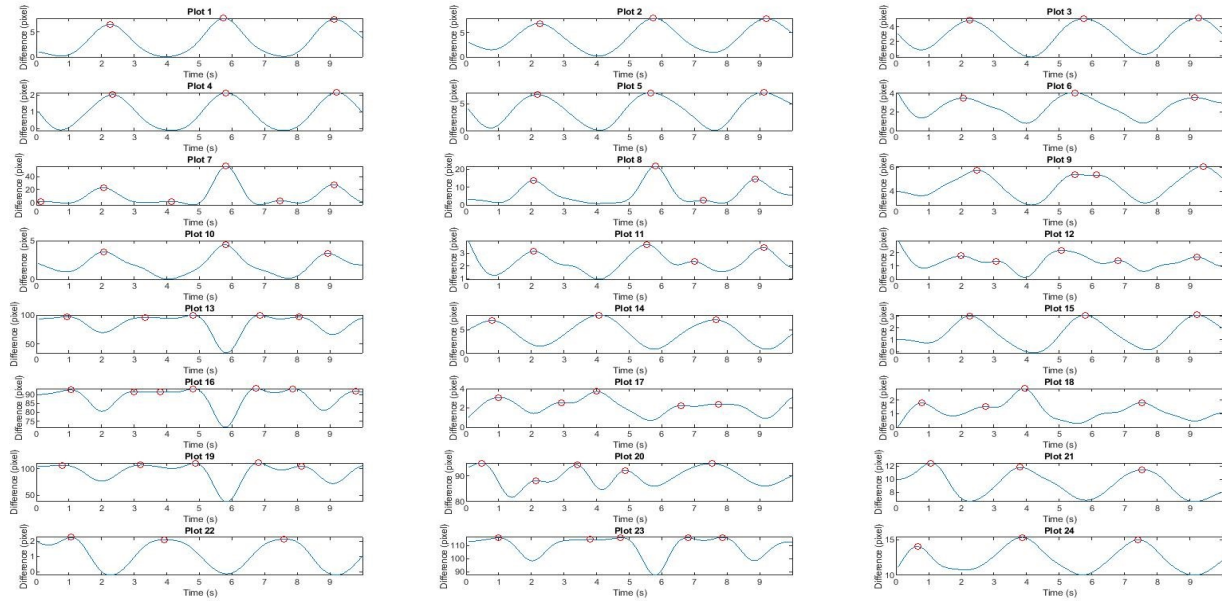


Fig. 9. Respiration signals for 10 seconds by 24 white dots

The correlation coefficient between the actual values and the experimental values was calculated 0.91.

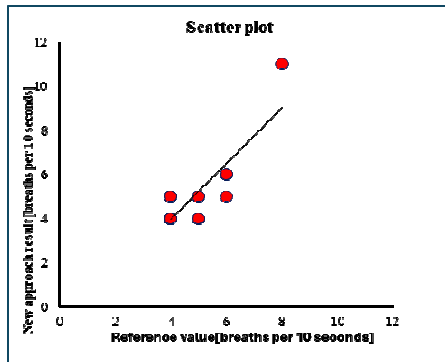


Fig. 10. Scatter plot of reference value and proposed method

IV. CONCLUSION

In this study, we proposed to develop a new approach for non-contact, rapid respiration rate measurement using a simple web camera and a mini projector. We generated a structured light pattern and illuminated the participants' chests. We enrolled 18 participants and recorded 10 seconds of each participant's video of when they usually breathed. Simultaneously, we counted participants' breaths manually by counting their chest movements. We calculated the correlation coefficient between the reference and the proposed method. The correlation coefficient was 0.91. The proposed method is promising for non-contact, rapid, and reliable respiration rate measurement based on the result. Because the method is non-contact, it can measure the respiration rate of newborn babies and elderlies. In future, we will develop a method to detect pneumonia.

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